

Passive FTC for Full-State Feedback and LQR

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1 Introduction

As we know, fault-tolerant control (FTC) is a method of estimating and accommodating various faults within the plant, sensors, actuators, etc. However, even within FTC there are more things we must consider, some of them being passive FTC versus active FTC.

Passive FTC is the capability of a control system to accommodate a fault without any dedicated online estimation or accommodation strategies. Purely by offline design of the controller, a fault can be accommodated to a certain degree.

Active FTC deals with control systems designed with dedicated fault estimation and accommodation steps built into the control algorithm. It is the most robust form of FTC, however it can be quite computationally costly compared to passive FTC.

Before we get into studying active control, it would be good to understand how certain control methodologies fare for passive control. We will explore full-state feedback using an arbitrary stable gain matrix K and compare it to LQR with varying values of Q and R , and see if the optimality of LQR does anything special to the stability of the system. The system we will be using is the two-state, non-isothermal CSTR system operating at the unstable equilibrium point.

2 The System

2.1 The Model

The state-space model we will be using is,

$$\dot{x} = Ax + B\theta u \tag{1}$$

$$\dot{\hat{x}} = \hat{A}\hat{x} + \hat{B}\hat{\theta}u \tag{2}$$

$$u = K\hat{x}, \tag{3}$$

which describes faults as multiplicative to the control input. The system we are studying with this state-space model is the two-state non-isothermal CSTR system centered around an unstable equilibrium point, where the manipulated variable is the volumetric inlet flow rate. The LTI state and input matrices for this system have already been calculated:

$$A = \begin{bmatrix} -4.578 \times 10^{-2} & -4.571 \times 10^{-4} \\ 2.063 \times 10^{-9} & 5.803 \times 10^{-2} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2.13 \\ -426.62534 \end{bmatrix}.$$

The steady-state inlet flow rate is $u_{eq} = F_{eq} = 3.34 \times 10^{-3} \text{ m}^3/\text{s}$. The equilibrium point is found to be,

$$x_{eq} = \begin{bmatrix} C_A^{eq} \\ T^{eq} \end{bmatrix} = \begin{bmatrix} 0.577 \text{ kmol/m}^3 \\ 395.262534 \text{ K} \end{bmatrix}.$$

We can confirm that A is centered around an unstable equilibrium point by checking its eigenvalues, which are $\lambda = [-0.0334 \ 0.04547603]$. Since one of them is negative and one is positive, we know that A is indeed centered around an unstable equilibrium. We are not required to study a system centered at an unstable equilibrium point, but it makes the dynamics, and thus the analysis, more interesting.

2.2 The Full-State Feedback Controller

A full-state feedback controller was studied that had the controller gain K_{FS} chosen as,

$$K_{FS} = [153.7797276 \ 0.77741062], \quad (4)$$

which places the closed-loop poles at $s = [-2.1 \ -2.0]$, making it stable.

2.3 The LQR Controller

The LQR controller was designed around $Q \in \mathbb{R}^{n \times n}$ and $R \in \mathbb{R}^{m \times m}$ matrices defined as,

$$Q = \frac{q}{x_{eq}^\top x_{eq}} I, \ x_{eq} \neq 0, \quad (5)$$

and,

$$R = \frac{r}{u_{eq}^\top u_{eq}} I, \ u_{eq} \neq 0, \quad (6)$$

where, to be as precise as possible, we define the squared inverse vectors as,

$$\frac{1}{x_{eq}^\top x_{eq}} = \text{diag} \left(\sum_i^n \frac{1}{x_{eq,i}^2} \vec{\delta}_i \right), \quad (7)$$

and,

$$\frac{1}{u_{eq}^\top u_{eq}} = \text{diag} \left(\sum_j^m \frac{1}{u_{eq,j}^2} \vec{\delta}_j \right). \quad (8)$$

The definitions for the squared inverse vectors may need some work but we can deal with that later. The parameters $q \geq 0$ and $r > 0$ are scalar variables that we can change to whatever we want. We choose to multiply by the squared inverse of the equilibrium states and inputs because this allows us to normalize the costs so that one state's cost doesn't inadvertently "overpower" another's, skewing the results.

2.4 The Augmented State

We can define an augmented state vector $\xi = [x^\top \ e^\top]^\top$, $e = x - \hat{x}$, which allows us to reform the state-space model as,

$$\dot{\xi} = \Lambda \xi, \quad (9)$$

where,

$$\Lambda = \begin{bmatrix} A + B\theta K & -B\theta K \\ \left((A - \hat{A}) + (B\theta - \hat{B}\hat{\theta}) K \right) & \hat{A} - (B\theta - \hat{B}\hat{\theta}) K \end{bmatrix} \quad (10)$$

This augmented state ξ and corresponding autonomous state matrix Λ allows us to describe the behavior of a sampled data system as an autonomous system, where the controller uses only estimates of the state and not the state itself.

As Λ describes the evolution of the continuous-time system, if all its eigenvalues are below 0 then the system is stable; if all of its eigenvalues are above 0 then the system is unstable.

2.5 The Sampled Data System

The stability of the sampled data system can be described through the expression,

$$M(t) = I_0 e^{\Lambda t}, \quad (11)$$

where $I_0 = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$ and I is the identity matrix of appropriate dimensions. As we are dealing with a sampled data system, t designates the period of time between samples, and in between samples the controller operates continuously on continuous estimations of the state. When the sampling time comes up, the state estimation is updated with a state measurement and the process repeats. We are interested specifically in a sampling period of Δ , so we focus on the stability of $M(\Delta)$. As $M(\Delta)$ now describes a discrete-time system, we require that all the eigenvalues of $M(\Delta)$ are below 1, and more specifically we care about the maximum eigenvalue, $\lambda_{max}(M(\Delta))$.

The name of the game is this: we will calculate $\lambda_{max}(M(\Delta))$ for various fault magnitudes $0 \leq \theta \leq 1$ using both the full-state feedback gain and the LQR gain and see which one has a larger stability region.

3 Results

3.1 $\lambda_{max}(M(\Delta))$ for Full-State Feedback and LQR

We can plot the values of $\lambda_{max}(M(\Delta))$ versus θ for both full-state feedback and LQR to see how they respond. Figure (1) shows that for this specific choice of K_{FS} , the system remains stable for more severe faults. It appears that the eigenvalues for LQR start to taper off for smaller faults, which may mean that LQR maintains a more “standard” level of stability for a wider range of fault—we may be able to rely on LQR more for behaving more consistently than full-state feedback, even if full-state feedback remains generally stable for a wider range of faults.

This more consistent level of stability is seen more clearly in figure (2), where the eigenvalues for full-state feedback still have the nearly linear behavior (except at the two ends of the domain), and LQR stays nearly constant for the entire range of faults.

3.2 The θ - Q and θ - R Parameter Spaces

Plotting $\lambda_{max}(M(\Delta))$ for varying q and r parameters within the θ - Q and θ - R parameter spaces yields quite interesting dynamics, as seen in figure (3). There do indeed appear to be “better options” for a given θ —for

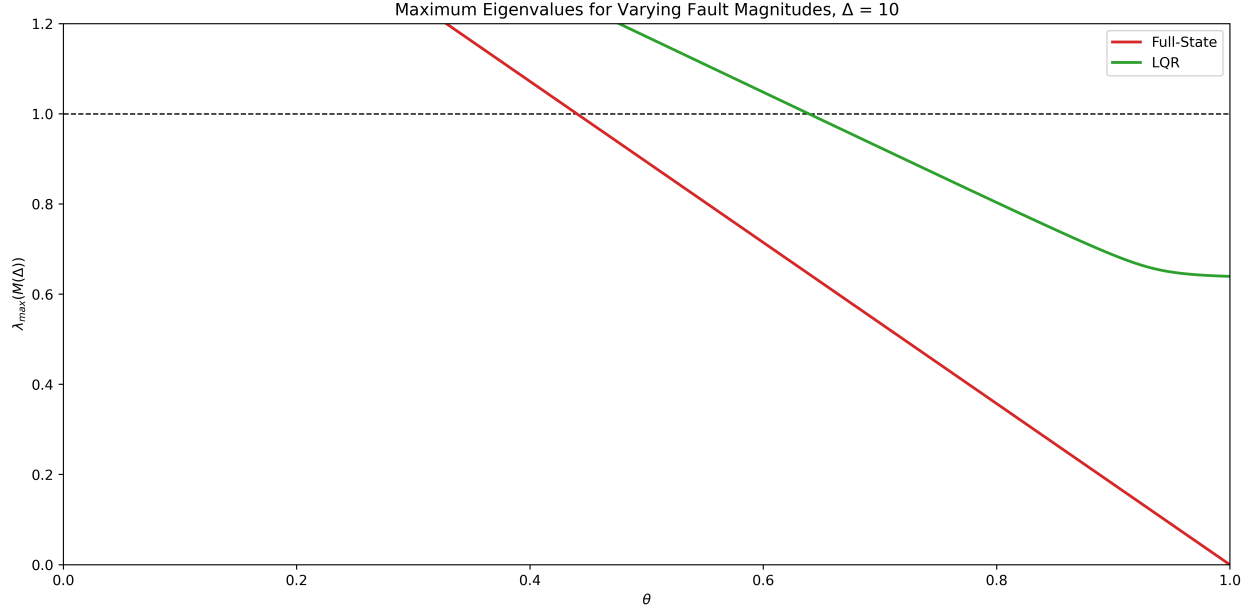


Figure 1: $\lambda_{\max}(M(\Delta))$ for full-state feedback and LQR, $\Delta = 10$; $Q = 1$ and $R = 1$. The dotted line indicates eigenvalues over 1, and anything above it will yield an unstable system.

example, if you have a minor fault of $\theta \geq 0.85$, then having a smaller Q appears favorable, however if you have a more severe fault of, say, $\theta = 0.5$, then having a very large Q would be better. Interestingly (but perhaps not so surprisingly), it almost seems to be the opposite for R —if you have a smaller fault, then having a larger R seems favorable, and if the fault is severe then having a smaller R is better.

These results make sense: if you have a small fault, then having a smaller Q would lead to a close-loop response that is, for lack of a better word, “flow-y,” and you allow the controller to apply smaller inputs to steer its way to the equilibrium point more smoothly; contrast this with a larger fault, where you’d want to penalize deviations from the equilibrium point because it is very likely that your system will veer away from the equilibrium point faster than the controller can account for. For the case of R , if the fault is small then you can afford to place large penalties on the input to achieve the “flow-y” response; contrasting with a large fault, you want to lessen the penalty on control inputs as the system may need larger inputs to counteract the fault.

We can verify that these results make sense by plotting the dynamic response of a system accommodated with a Q that lies within the stable region, as well as another system accommodated with an unstable Q . This is seen in figure (4). The same was done for R , shown in figure (5).

3.3 Δ Dependence

It is not surprising that given a plant-model mismatch through θ , if the sampling time Δ is large then the system becomes more unstable—after all, if you’re operating for a long time based on an inaccurate estimate, then it is extremely unlikely that you’d be controlling the actual system to a suitable degree of efficacy.

Let’s do the same analysis as we did for the $\Delta = 1$ case, where this time we increase it to $\Delta = 20$ to start really seeing some interesting dynamics. Figure (6) shows the new contours for the stability/instability regions, and we see that our range of faults we can accommodate has drastically decreased. Pretty much all

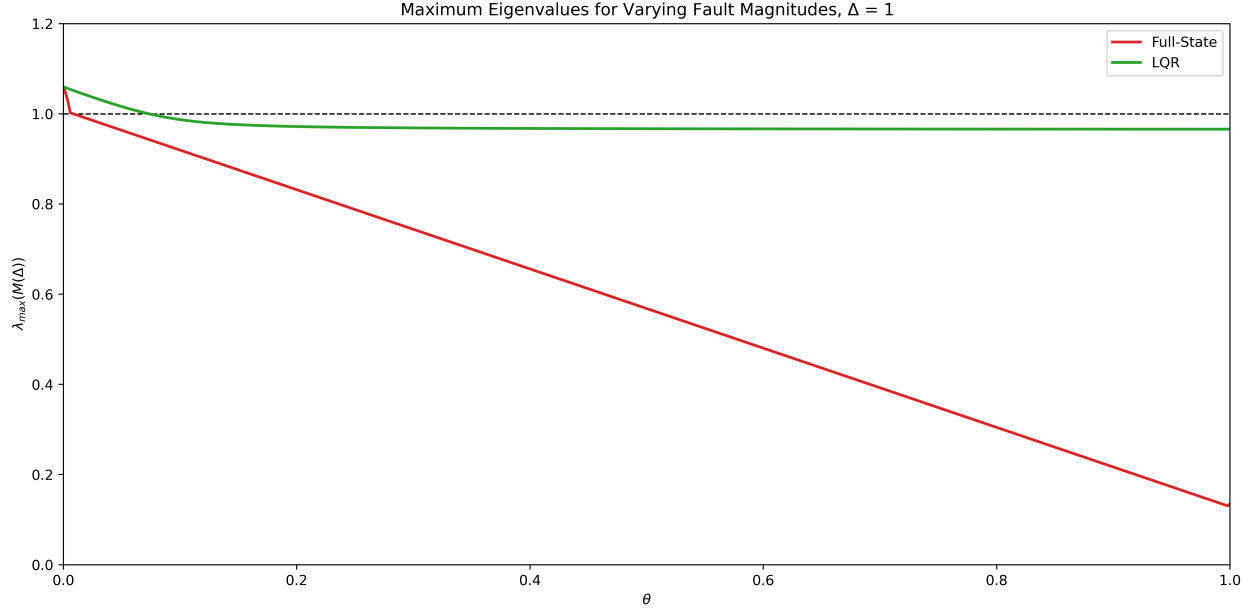


Figure 2: $\lambda_{max}(M(\Delta))$ for full-state feedback and LQR, $\Delta = 1$; $Q = 1 \times 10^5$ and $R = 1$. The dotted line indicates eigenvalues over 1, and anything above it will yield an unstable system.

faults more severe than about $\theta = 0.7$ are unable to be accommodated without redesigning the controller logic.

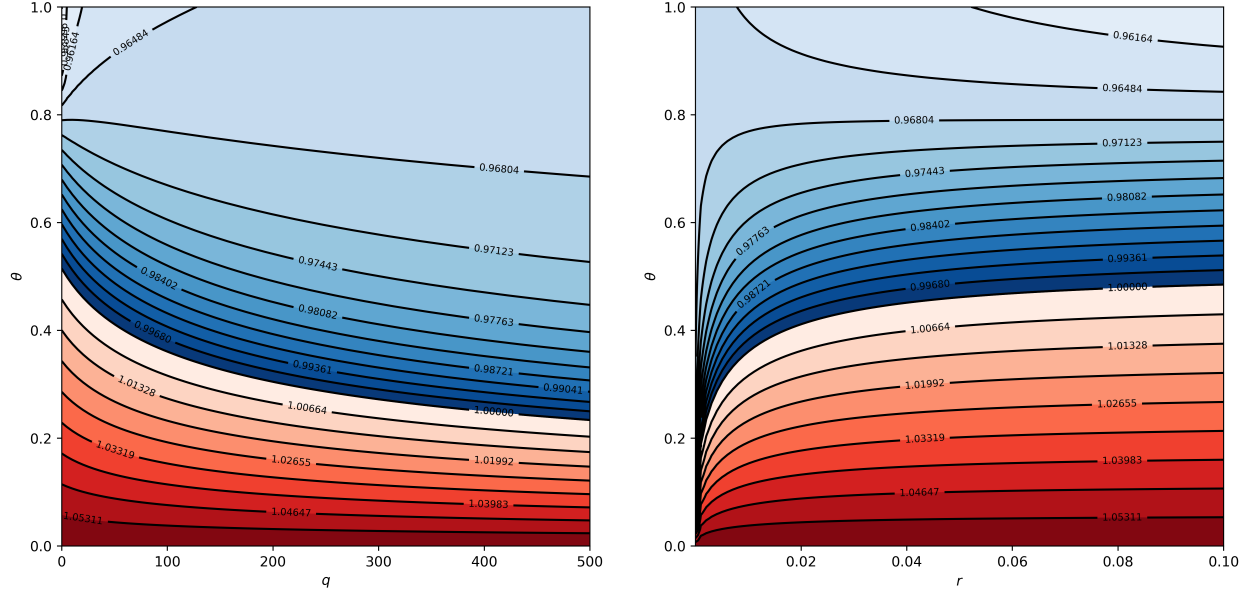
Again we can verify that these contours are correct by simulating a dynamic model with stable and unstable choices of Q and R . Figure (7) shows the affect of a large Δ with stable and unstable Q s, where the larger sampling time allows us to more easily see how the system is behaving. The red dashed lines are the state estimate, which between samples believes that it is controlling the system adequately and steering it back to the equilibrium point, but the actual states, in black, not only are not responding as strongly as the estimates, but are veering off from the course because of the unstable equilibrium point. The choice of Q here is strong enough that between sampling times, the controller pulls on the states strong enough to overcome the instability around the equilibrium, eventually leading to an overall stable system. The same cannot be said about the unstable choice of Q , where the controller is not pulling as hard to overcome the instability about the equilibrium, so the states slowly begin to diverge.

Figure (8) shows the same but for R . The story is similar, but again with the slight change in perspective where this time we're penalizing control inputs more strongly. With a small enough R , we allow the controller to apply a stronger action and it is able to overcome the instability, but with a large R we penalize the input so much that the controller is not able to strong-arm its way out of instability.

Based off of these results, we can say that increasing the sampling time Δ will decrease the size of this stability region, and vice versa. As this is an unbounded and unconstrained control problem, there really shouldn't be any stable or unstable regions within state-space, since the controller *should* be able to apply as much control as it needs whenever it needs it, as really the cost function is sort of balancing Q vs. R and not x vs. u .

We can further see how the range of acceptable faults changes as Δ changes by plotting $\lambda_{max}(M(\Delta))$ contours in θ - Δ space, and see how different values of Q and R further change the shape. We will create many contour plots to visualize this. First up we have θ - Δ contours while varying Q , shown in figure (9).

$\lambda_{\max}(M(\Delta))$ with Varying Q and R , $\Delta = 1$



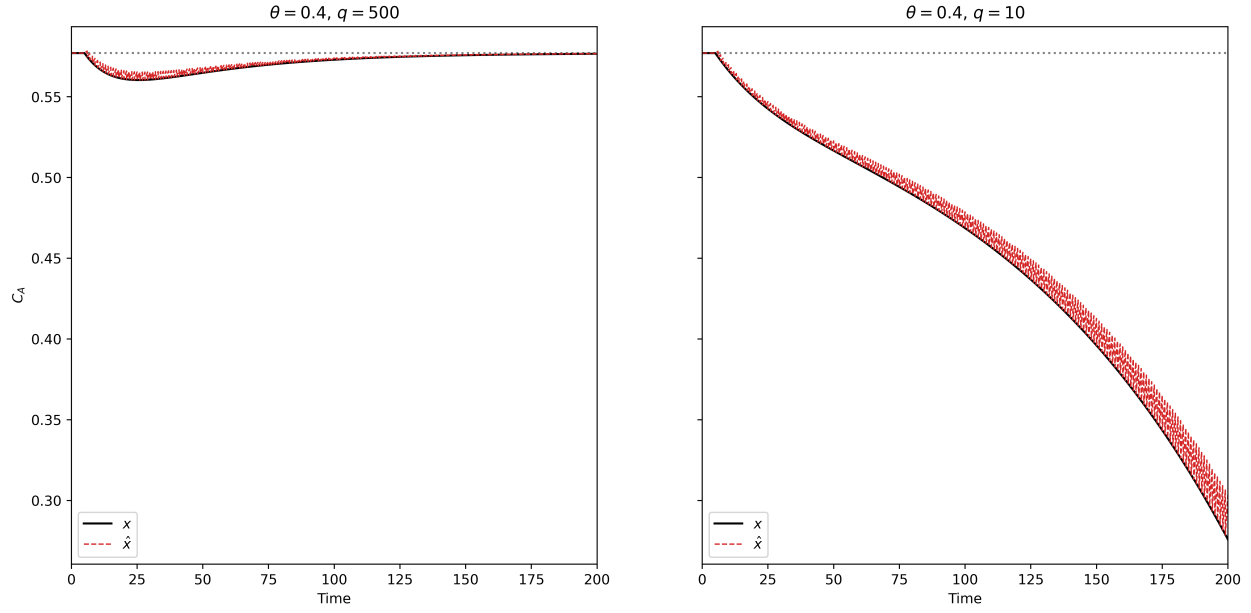


Figure 4: Dynamic responses of an LQR-controlled system with a stable choice of Q (left) and an unstable choice of Q (right). These results verify that the contour lines are correct.

makes it harder for you to miss key information about its behavior. These results also provide a general guide for understanding what faults you can get away with for a given LQR controller. The next thing I want to investigate is how LQR

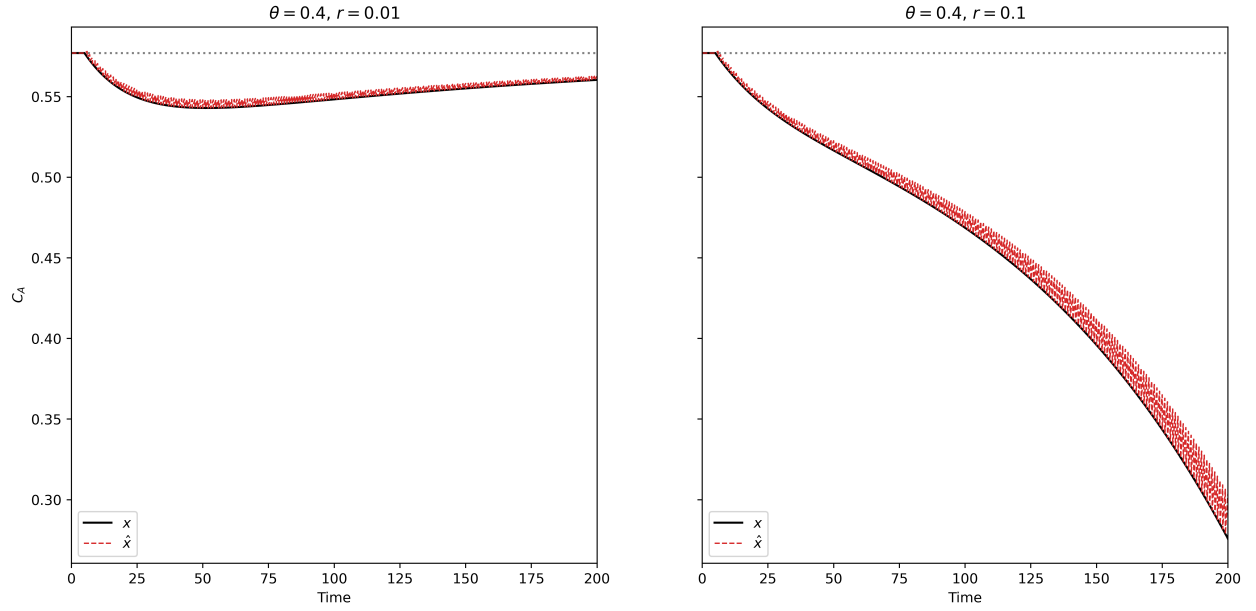


Figure 5: Dynamic responses of an LQR-controlled system with a stable choice of R (left) and an unstable choice of R (right). These results also verify that the contour lines are correct.

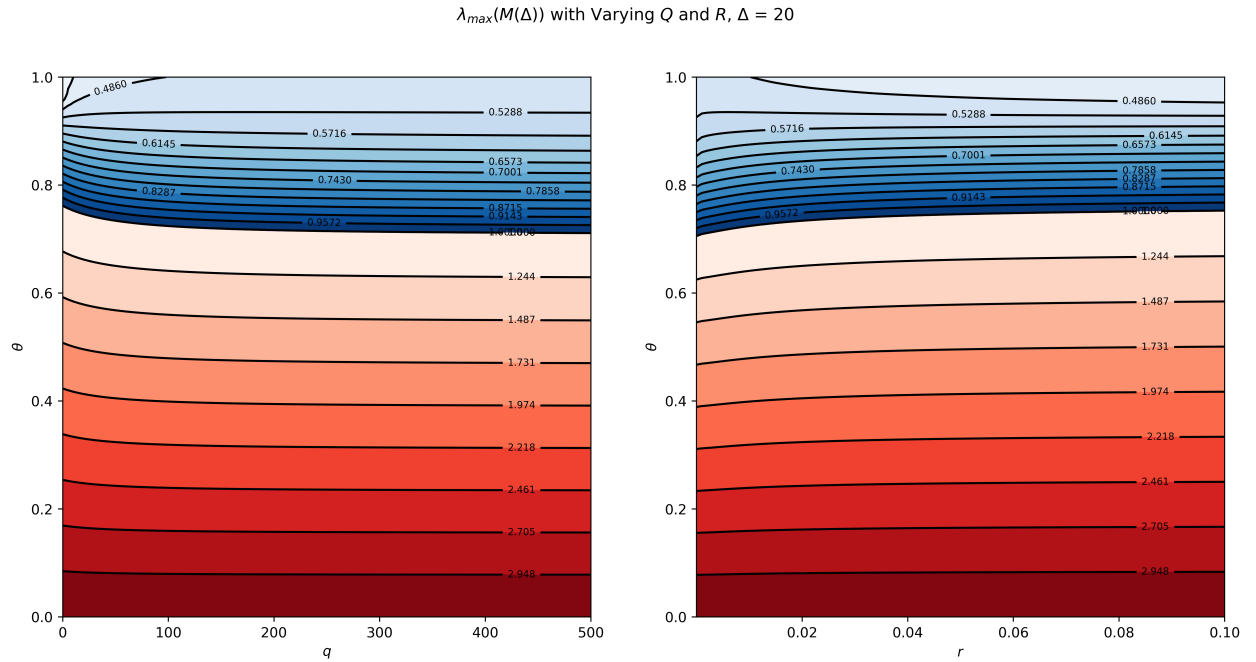


Figure 6: Contours of $\lambda_{max}(M(\Delta))$, $\Delta = 20$. The blue region is the stable region in each parameter space, and the red region is the instability region.

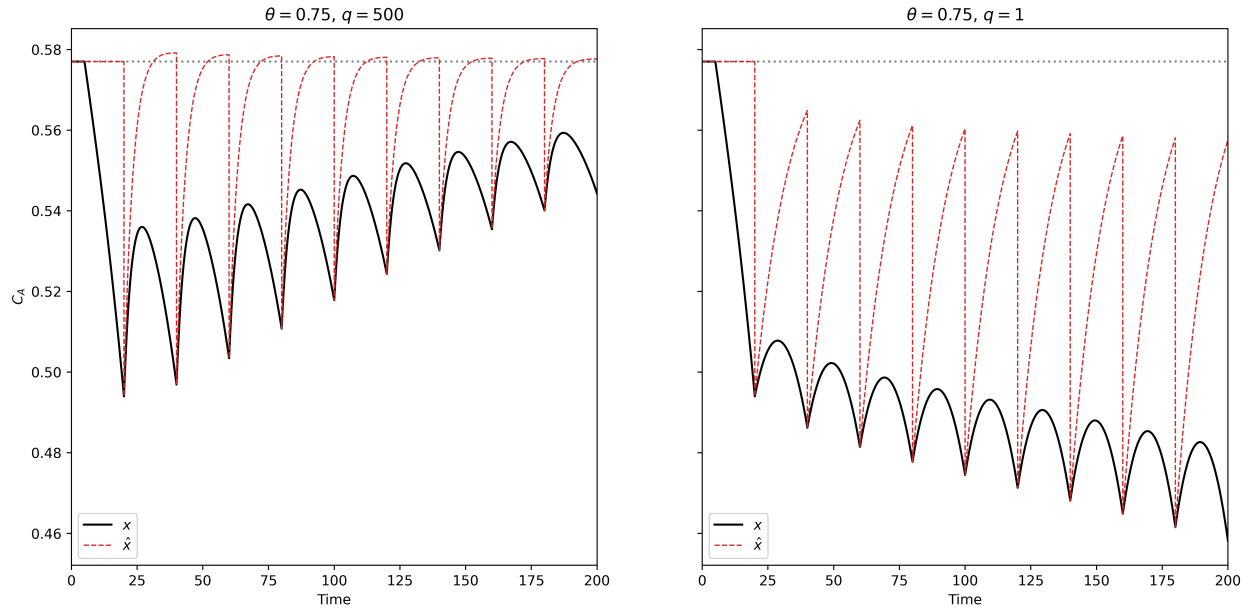


Figure 7: Dynamic responses of an LQR-controlled system with a stable choice of Q (left) and an unstable choice of Q (right). These results verify that the contour lines are still correct.

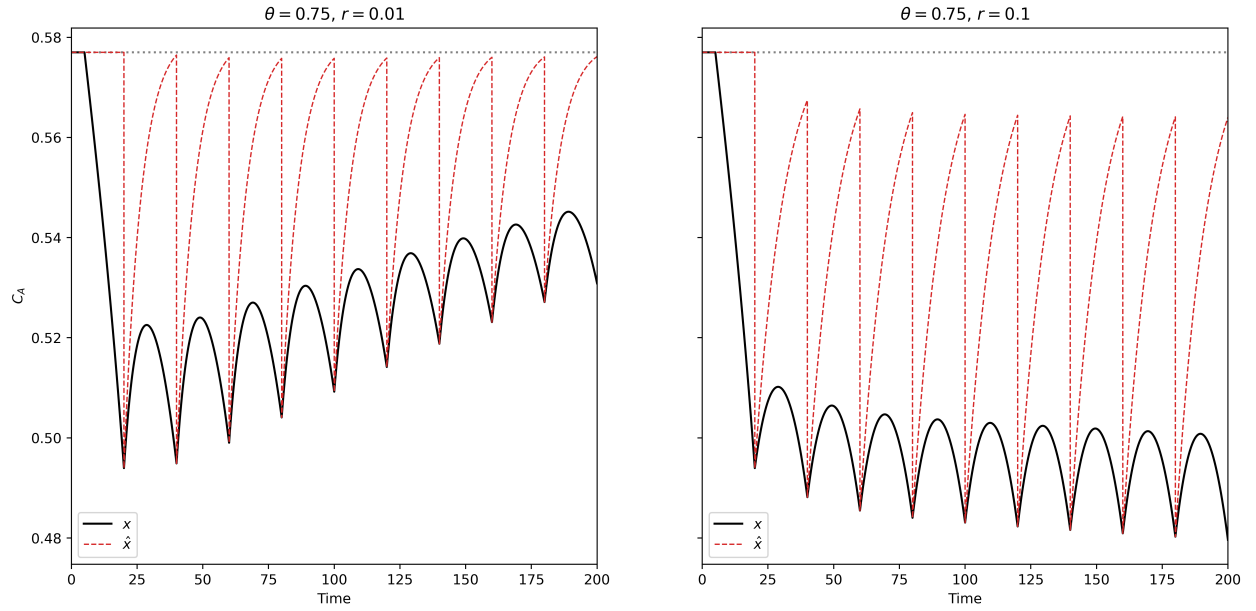


Figure 8: Dynamic responses of an LQR-controlled system with a stable choice of R (left) and an unstable choice of R (right). These results also verify that the contour lines are still correct.

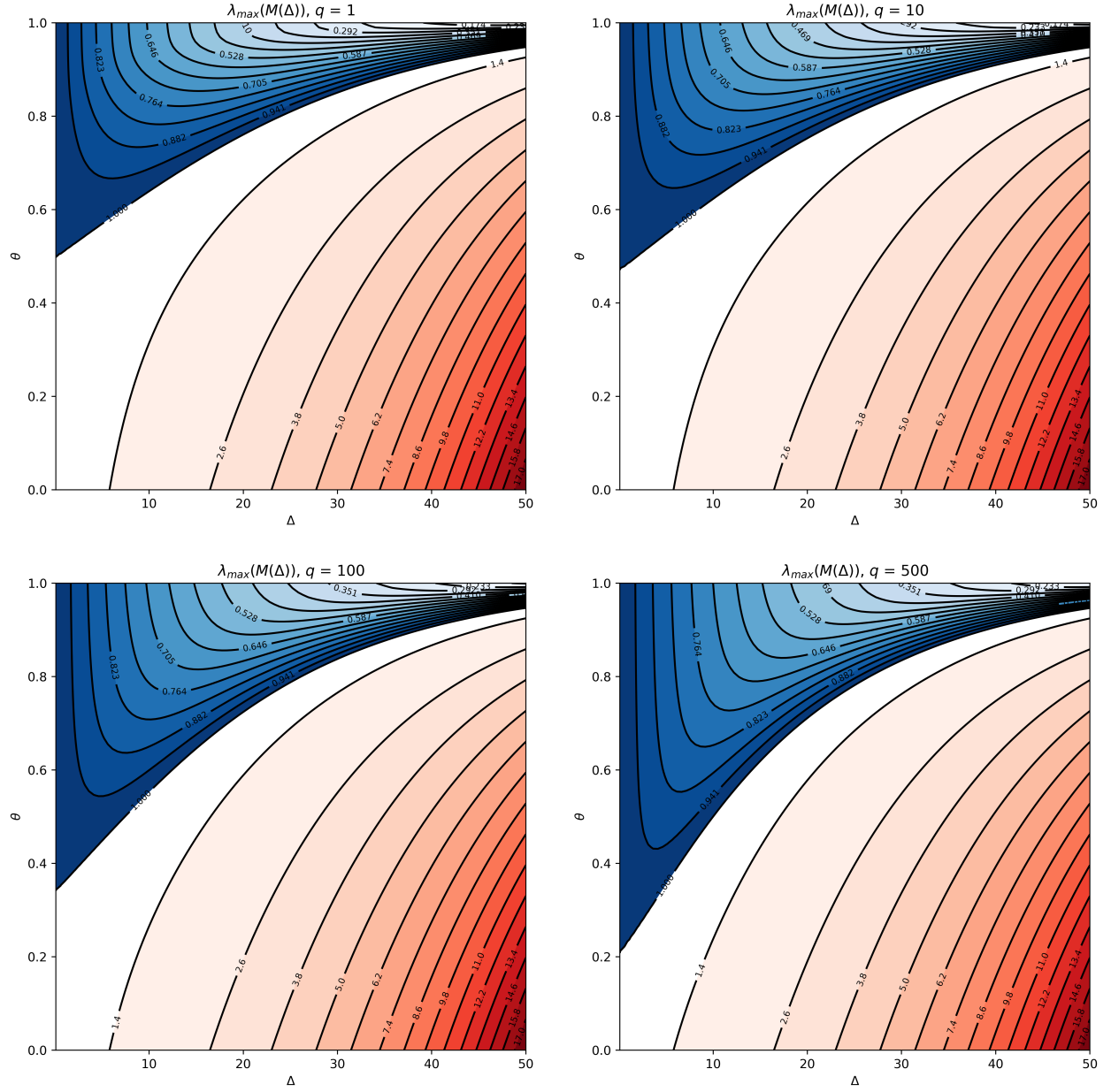
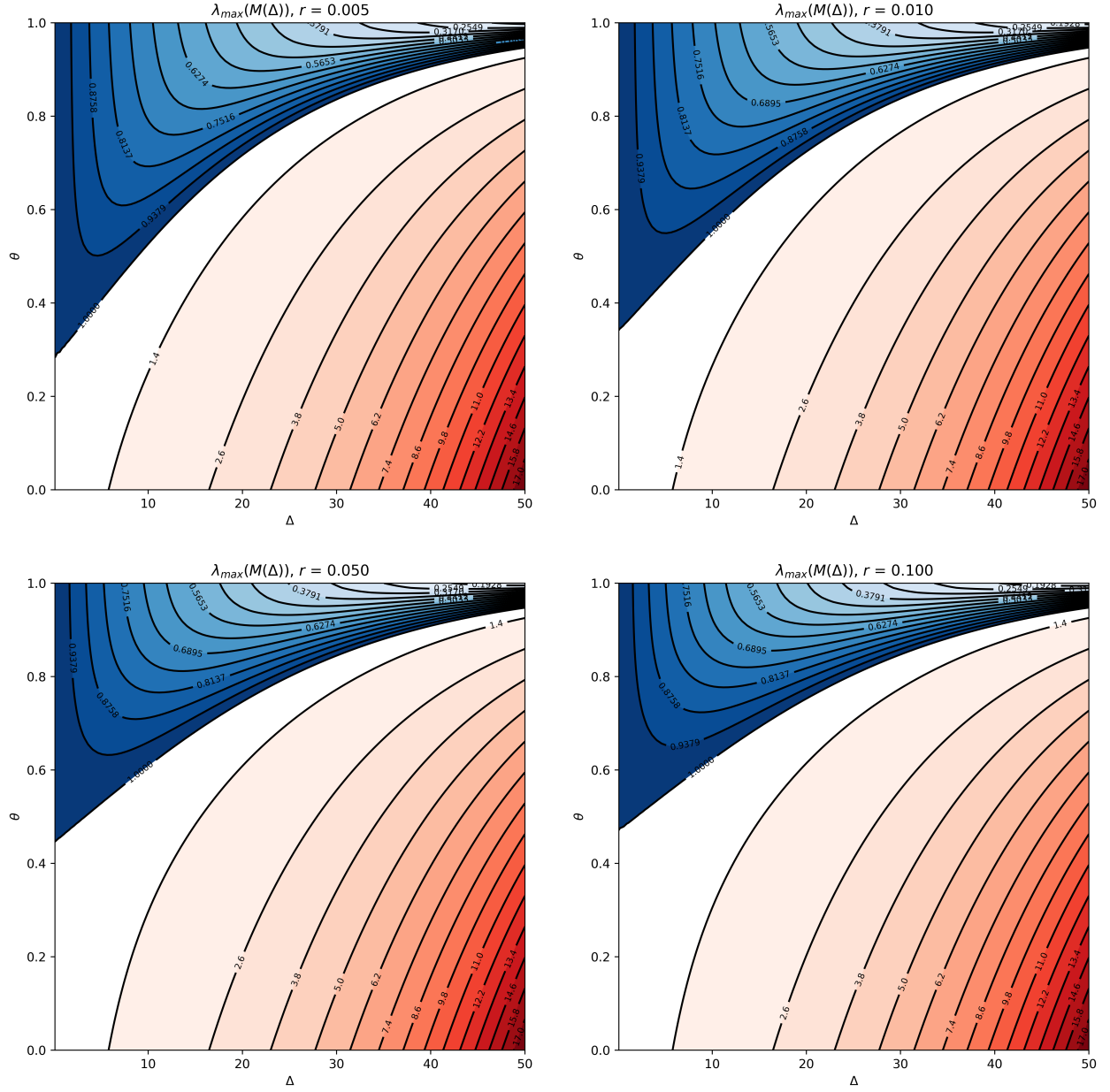


Figure 9: Contours of $\lambda_{\max}(M(\Delta))$ in θ - Δ parameter space, varying the value of Q . The blue region is the stable region in each parameter space, and the red region is the unstable region.



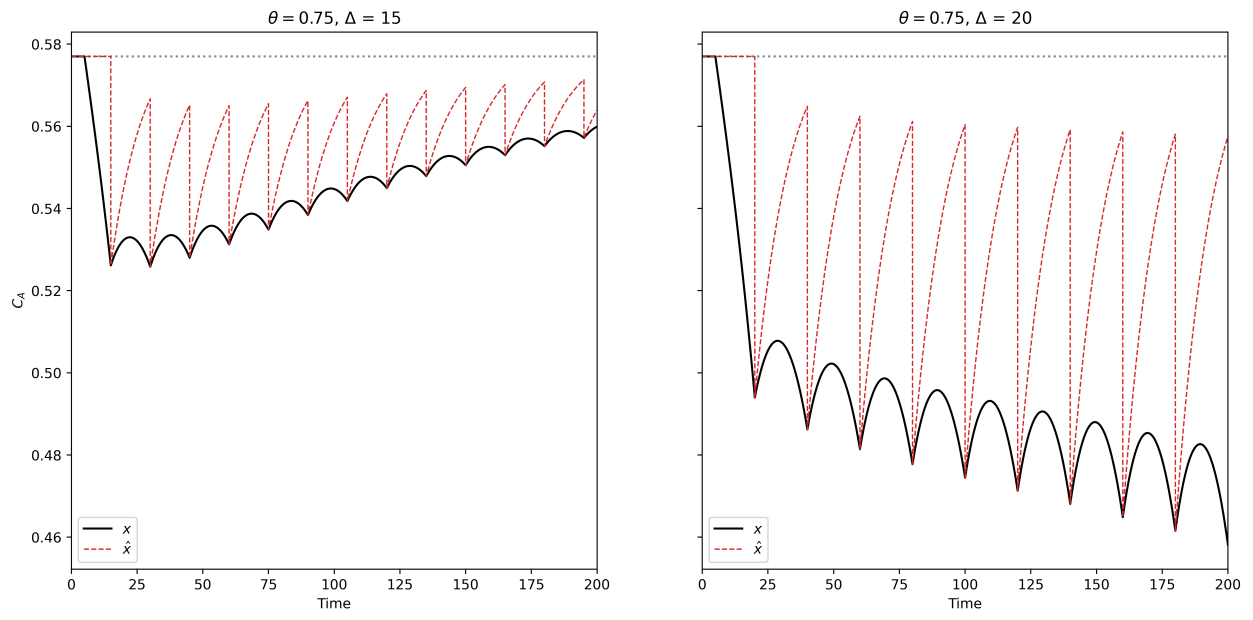


Figure 11: Dynamic responses of an LQR-controlled system with a stable choice of Δ (left) and an unstable choice of Δ (right). These results verify that the contour lines are correct.